MIL-STD-1639A
22 February 1980
SUPERSEDING
MIL-STD-1639
6 January 1978

MILITARY STANDARD

POWER DIVIDERS, POWER COMBINERS, AND POWER DIVIDER/COMBINERS, SELECTION OF

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Power Dividers, Power Combiners, and Power Divider/Combiners, Selection of.

MIL-STD-1639

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ELEX 5043, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-24T00:48Z
Check the source to verify that this is the current version before use.

CONTENTS

		PAGE
Paragraph 1.	SCOPE	1
1.1	Scope	1
1.2	Purpose	1
2.	REFERENCED DOCUMENT	1
2.1	Issues of document	1
3.	DEFINITIONS	1
4.	GENERAL REQUIREMENTS	1
4.1	Selection of power dividers or combiners	1
4.2	Criteria for selection	1
4.3	Application and use	1
5.	DETAILED REQUIREMENTS	2

TABLE

Table I.	Part numbers and characteristics	2-6

Source: https://assist.dla.mil -- Downloaded: 202610-24T00:48Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard provides standard power dividers, power combiners, and power divider/combiners considered by the Department of Defense as standard for use in military equipment and applications.

1.2 Purpose. The purpose of this standard is to:

a. Provide the equipment designer with a list of power dividers or combiners considered standard for use in military applications.
b. Restrict and minimize the variety of power dividers or combiners for use in military applications in order to provide effective logistic support of equipment.
c. Establish criteria pertinent to choice and application of power dividers or combiners for use in military equipment.

2. REFERENCED DOCUMENT

2.1 Issues of document. The following document of the issue in effect on the date of invitation for bids or request for proposal forms a part of this standard to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-23971 - Power Dividers, Power Combiners, and Power Divider/Combiners, General Specification For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 The terms used in this standard are those commonly encountered in microwave engineering practice.

4. GENERAL REQUIREMENTS

4.1 Selection of power dividers or combiners. Power dividers or combiners to be used in military applications shall be selected from those listed in Table I.

4.2 Criteria for selection. The criteria for the selection of power dividers or combiners for inclusion in this standard are:

a. The power dividers or combiners shall be considered by representatives of the military departments to be the best available type for current application.
b. Availability of the power dividers or combiners shall be reasonably certain.
c. The power dividers or combiners shall have an approved military specification.

4.3 Application and use. Power dividers or combiners used in military application shall be representative of manufactured lots possessing acceptable material and physical and electrical characteristics and shall in no manner degrade the operational characteristics of the equipment in which used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 The detailed requirements for power dividers or combiners listed in this standard are covered by MIL-P-23971 specification sheets.

TABLE I. Part numbers and characteristics.

Part number	Frequency range	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance degrees	Amplitude balance (dB)	Power level (W)	
								Avg	Pk
POWER DIVIDERS, 4 PORT QUADRATURE, SOLDER TERMINALS									
M23971/1-01	225-400 MHz	3+.2 -0	1.2:1	0.3	23	±1.5	±0.4	60	300
-02	250-500 MHz	3+.2 -0	1.2:1	0.25	23	±1.5	±0.5	60	300
-03	225-400 MHz	3+.2 -0	1.2:1	0.25	25	±1.5	±0.4	100	200
-04	225-400 MHz	3+.2 -0	1.2:1	0.25	25	±1.5	±0.4	100	200
POWER DIVIDERS, 4 PORT QUADRATURE, SMA CONNECTORS 2/									
M23971/2-01	0.25-1.0 GHz	3+.2 -0	1.3:1	0.5	20	±1.5	±0.75	100	5K
-02	1.0-2.0 GHz	3+.2 -0	1.3:1	0.25	23	±1.5	±0.5	100	5K
-03	2.0-4.0 GHz	3+.2 -0	1.3:1	0.25	20	±1.5	±0.5	100	5K
-04	4.0-8.0 GHz	3+.2 -0	1.35:1	0.35	18	±1.5	±0.5	100	5K
-05	0.02-.200 GHz	3+.5 -0	1.3:1	1.3	20	±5	±1.0	1	–
-06	0.5-1.0 GHz	3+.2 -0	1.25:1	0.25	18	±1.5	±0.5	50	250
-07	1.0-2.0 GHz	3+.2 -0	1.25:1	0.25	18	±1.5	±0.5	50	250
-08	2.0-4.0 GHz	3+.2 -0	1.25:1	0.25	18	±2.0	±0.5	50	250
-09	4.0-8.0 GHz	3+.2 -0	1.25:1	0.35	18	±2.0	±0.5	50	250
-10	7.0-11.0 GHz	3+.5 -0	1.45:1	0.50	17	±2.0	±0.5	100	5K
-11	8.0-12.4 GHz	3+.2 -0	1.30:1	0.40	15	±3.0	±0.5	50	250
-12	12.4-18.0 GHz	3+.2 -0	1.35:1	0.50	15	±5.0	±0.5	50	250
POWER DIVIDERS, 4 PORT QUADRATURE, PLUG-IN TERMINALS									
M23971/3-01	225-400 MHz	3+.5 -0	1.2:1	0.25	25	±1.5	0.4	100	200
POWER DIVIDERS, 3 PORT QUADRATURE, SOLDER TERMINALS									
M23971/4-01	225-400 MHz	3+.2 -0	1.2:1	0.25	20	±1.5	0.4	100	200

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics – Continued.

Part number	Frequency range	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance degrees	Amplitude balance (dB)	Power level (W)	
								Avg	Pk
POWER DIVIDERS, N-WAY, 0-DEGREES, SOLDER TERMINALS									
M23971/5-01	5-25 MHz	3 +.2 −0	1.3:1	0.3	25	±1.0	±0.1	0.5	1.0
-02	0.2-200 MHz	4.8 +.5 −0	1.3:1	0.5	25	±2.0	±0.2	0.5	0.5
-03	10-500 MHz	3+.2 −0	1.3:1	0.5	30	±1.0	±0.1	1	–
POWER DIVIDERS, N-WAY, 0-DEGREES, PLUG-IN TERMINALS									
M23971/6-01	1.0-100 MHz	3 +.5 −0	1.3:1	0.6	30	±1.0	±0.1	.5	–
-02	2-32 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	.1
-03	2-100 MHz	3 +.2 −0	1.2:1	0.25	40	±1.0	±0.1	–	3
-04	10-100 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	.1
-05	20-200 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	.1
-06	100-300 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	.1
-07	200-400 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	.1
-08	10-100 MHz	4.8 +.3 −0	1.3:1	0.5	30	±1.0	±0.3	1	–
-09	10-200 MHz	4.8 +.3 −0	1.3:1	0.5	30	±2.0	±0.2	–	3
POWER DIVIDERS, 4 PORT QUADRATURE, N CONNECTORS									
M23971/7-01	7.0-12.4 GHz	3 +.2 −0	1.35:1	0.1	15	±1.5	±0.5	100	3K
POWER DIVIDERS, QUADRATURE, FLAT PACK									
M23971/8-01	7.0-14.0 MHz	3 +.2 −0	1.2:1	0.5	25	±2.0	±0.75	5	–
M23971/8-02	40-80 MHz	3 +.2 −0	1.2:1	0.5	20	±3.0	±0.75	5	–
POWER DIVIDERS, N-WAY, 0-DEGREES, SMA CONNECTORS									
M23971/9-01	0.5-1.5 GHz	3 ±.25	1.15:1	0.05	6	±6.0	±1.0	150	2K

Source: https://assist.dla.mil -- Downloaded: 202610-24T00:48Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics – Continued.

Part number	Frequency range	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance degrees	Amplitude balance (dB)	Power level (W)	
								Avg	Pk
POWER DIVIDER/COMBINERS, QUADRATURE, RECTANGULAR, PC PINS									
M23971/10-02	26-34 MHz	3 +.2 −0	1.2:1	0.3	25	±1.0	±0.2	1	–
POWER DIVIDER/COMBINER, QUADRATURE, TO-5									
M23971/11-01	29-31 MHz	3 +.5 −0	1.2:1	0.2	25	±2.0	±0.6	–	.25
M23971/11-02	91.2-100.8 MHz	3 +.2 −0	1.2:1	0.2	25	±2.0	±0.6	–	.25
POWER DIVIDER/COMBINERS, QUADRATURE, STRIPLINE									
M23971/12-01	0.1-0.16 GHz	3 +.2 −0	1.2:1	0.3	25	±2.0	±0.6	200	1K
M23971/12-02	0.5-1.0 GHz	3 +.2 −0	1.2:1	0.25	20	±1.5	±0.5	200	1K
POWER DIVIDER/COMBINERS, 2-WAY, 180 DEGREES, SMA CONNECTORS									
M23971/13-01	1.0-2.0 GHz	3 +.2 −0	1.35:1	0.5	20	±6.0	±0.2	50	4K
M23971/13-02	2.0-4.0 GHz	3 +.2 −0	1.5:1	0.65	20	±6.0	±0.4	50	4K
M23971/13-03	4.0-8.0 GHz	3 +.2 −0	1.6:1	0.6	15	±8.0	±0.5	50	4K
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, FLAT PACK									
M23971/14-01	5.0-100 MHz	3 +.2 −0	1.3:1	0.4	30	±1.0	±0.1	–	1.0
M23971/14-02	10-500 MHz	3 +.6 −0	1.3:1	0.6	30	±1.0	±0.1	–	1.0
M23971/14-03	300-500 MHz	3 +.2 −0	1.3:1	0.5	30	±2.0	±0.1	–	1.0
M23971/14-04	3.0-32 MHz	4.8 +.3 −0	1.2:1	0.6	30	±2.0	±0.2	–	0.5
M23971/14-05	10-500 MHz	6 +.85 −0	1.5:1	0.85	25	±2.0	±2.0	–	1.0
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, RECTANGULAR, PC PINS									
M23971/15-01	0.1-400 MHz	3 +1.0 −0	1.2:1	1.0	15	±4.0	±0.3	–	1
-02	0.1-450 MHz	3 +1.0 −0	1.2:1	0.6	15	±4.0	±0.3	–	1
-03	2.0-32 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	1
-04	10-100 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	1

Source: https://assist.dla.mil -- Downloaded: 202610-24T00:48Z
Check the source to verify that this is the current version before use.

TABLE I. Part numbers and characteristics – Continued.

Part number	Frequency range	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance degrees	Amplitude balance (dB)	Power level (W)	
								Avg	Pk
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, RECTANGULAR, PC PINS – Continued									
M23971/15-05	20-200 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	1
-06	50-1000 MHz	3 +.2 −0	1.3:1	1.0	25	±2.0	±0.3	–	1
-07	100-300 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	1
-08	200-400 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	1
-09	140-410 MHz	4.8 +.3 −0	1.3:1	0.75	25	±3.0	±0.2	–	3
-10	0.1-200 MHz	6 +.4 −0	1.2:1	1.0	20	±2.0	±0.25	–	1
-11	0.25-250 MHz	6 +.75 −0	1.1:1	1.2	20	±2.0	±0.25	–	1
-12	10-100 MHz	6 +.4 −0	1.3:1	0.7	30	±1.0	±0.2	–	2
-13	235-275 MHz	7.78 +.6 −0	1.15:1	1.0	30	±2.0	±0.2	–	5
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, SMA CONNECTORS									
M23971/16-01	1.0-100 MHz	3.01 +.5 −0	1.3:1	0.5	30	±1.0	±0.2	–	2
-02	1.0-2.0 GHz	3 +.2 −0	1.25:1	0.3	20	±6.0	±0.4	–	.1
-03	1.0-2.0 GHz	3 +.2 −0	1.25:1	0.3	20	±6.0	±0.4	–	.1
-04	1.0-12.4 GHz	3 +.2 −0	1.7:1	1.3	16	±5.0	±0.2	1	–
-05	10-100 MHz	4.77 +.5 −0	1.3:1	0.5	30	±1.0	±0.2	–	5
-06	2.0-32 MHz	6 +.5 −0	1.3:1	0.7	30	±1.0	±0.2	–	2
-07	10-100 MHz	6.02 +.7 −0	1.3:1	0.7	30	±1.0	±0.2	–	2
-08	20-200 MHz	6 +.5 −0	1.3:1	0.7	30	±1.0	±0.2	–	2
-09	100-300 MHz	6 +.5 −0	1.3:1	1.0	30	±1.0	±0.2	–	2
-10	200-500 MHz	6 +.5 −0	1.3:1	1.0	30	±1.0	±0.2	–	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

TABLE I. Part numbers and characteristics – Continued.

Part number	Frequency range	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance degrees	Amplitude balance (dB)	Power level (W)	
								Avg	Pk
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, SMA CONNECTORS – Continued									
M23971/16-11	250-500 MHz	6 +.5 −0	1.3:1	1.0	30	±1.0	±0.2	–	2
-12	1.0-2.0 GHz	6 +.5 −0	1.35:1	0.6	20	±6.0	±0.6	–	.1
-13	1.0-2.0 GHz	6 +.9 −0	1.35:1	0.6	20	±6.0	±0.6	.316	–
-14	2.0-4.0 GHz	6 +.5 −0	1.35:1	0.6	18	±3.0	±0.3	.1	–
-15	10-100 MHz	7.8 +.7 −0	1.3:1	1.0	30	±2.0	±0.2	2	–
POWER DIVIDER/COMBINERS, N-WAY, 0-DEGREES, TNC CONNECTORS									
M23971/17-01	0.05-20 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	2
M23971/17-02	1.0-100 MHz	3 +.2 −0	1.3:1	0.5	30	±1.0	±0.2	–	5
M23971/17-03	200-400 MHz	4.8 +.3 −0	1.3:1	0.75	25	±2.0	±0.2	–	5

Custodians:
Army – ER
Navy – EC
Air Force – 11

Review activities:
Army – MI, AR
Navy – OS, SH
Air Force – 17, 85
DLA – ES

User activities:
Army – AV
Navy – AS, CG, MC
Air Force – 19

Preparing activity:
Navy – EC

Agent:
DLA – ES

(Project 5985-0924)

Source: https://assist.dla.mil Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-STD 1639A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:48Z
Check the source to verify that this is the current version before use.